Malhar Jayant Falke

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Education

COEP Technological University (College of Engineering, Pune) Pune, India
B.Tech, Computer Engineering; CGPA 6.83/10 *2023 – Expected May 2027*

- **Core:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Computer Architecture, Theory of Computation, Parallel Computing, Artificial Intelligence.
- **Systems electives:** Embedded Systems, Real-Time Operating Systems, Microcontrollers & Microprocessors, Digital Electronics, Sensor Interfacing & Communication Protocols.

Class XII	DDM English High School, Maharashtra State Board (MSBSHSE)	65.33%	2023
Class X	Vidya Niketan School, Maharashtra State Board (MSBSHSE)	76.00%	2021

Experience

Klimaforge Norway (Remote)
Machine Learning & Data Science Intern: geospatial machine learning *May 2026 – Present*

- Built an above-ground forest biomass (AGB) model for Estonia on ∼678k samples and 48 engineered features fused from Sentinel-1 SAR, Sentinel-2 multispectral and terrain (DTM: elevation, slope, aspect) data, with seasonal composites.
- XGBoost reached R^2 0.61 (RMSE 28.7 t/ha) on a held-out split and 0.58 in cross-validation; the deployed model produced the national 2023 AGB map. A later 31-feature fusion model reached cross-validated R^2 0.64 on 658k samples.
- Re-evaluated under 10 km spatial-block cross-validation (R^2 0.50) to measure how much random splits overstate accuracy under spatial autocorrelation; this became the reported benchmark.
- Ran sensor ablations under spatial blocking (Sentinel-1 only 0.32, Sentinel-2 only 0.47, S1+S2 0.52, S1+S2+ALOS L-band 0.55), showing each sensor adds signal and none replaces the others.
- Benchmarked the Prithvi-EO-2.0 geospatial foundation model, both fine-tuned and as frozen embeddings; it underperformed engineered features at every training size, so the classical model shipped.
- Caught and discarded a canopy-height feature that leaked the target (R^2 1.00), and tested a log-target variant (R^2 0.53) that was not deployed.
- Automated satellite data acquisition and preprocessing with Google Earth Engine, the Copernicus Data Space Ecosystem (CDSE) and openEO.

COEP Technological University Pune, India
Embedded Systems Research Fellow: firmware and real-time systems, under faculty supervision *Jun 2025 – Aug 2025*

- Implemented SPI, I^2C, UART, Bluetooth and Zigbee on PIC and ARM7/Cortex-M microcontroller targets.
- Wrote firmware for peripheral interfacing: graphical LCD, SPI TFT LCD and PWM-based motor control.
- Studied and implemented the FreeRTOS and μC/OS-II real-time operating systems.
- Authored self-learning modules (print and video) for Embedded C and Embedded Linux, and prepared lesson plans of tested experiments for PIC controllers.

Projects

AarthFit: MSME Financial Fitness Platform *Apr 2026 – Present*
Fastify, Node.js 22, TypeScript, React (Vite), PostgreSQL/TimescaleDB, Redis, Knex, AWS SES, pnpm

- Full-stack, multi-tenant SaaS built for a business consultancy in Dombivli, Maharashtra, to improve financial discipline and compliance tracking for small-business owners.
- Owners answer a daily question over WhatsApp; answers roll up into a CFO Score (cashflow, compliance, profitability, balance sheet, governance), benchmarked by business type and revenue tier and tracked on a self-serve dashboard.
- Designed the data architecture, REST API and relational schema: Fastify + TypeScript API in a pnpm monorepo with shared zod schemas, Knex migrations and repository functions with audit logging.
- Implemented Redis-backed OTP and JWT authentication with role-based access control; integrated Razorpay payments with webhook signature verification and the Meta WhatsApp Cloud API with template linting.
- Built a role-based admin portal (deals, companies, staff, accountant workspace) and a super-admin platform-metrics dashboard across tenants, using indexed, time-scoped aggregate queries behind a short Redis cache.
- Generated CFO Score PDF reports with pdfkit and delivered them over AWS SES; background jobs run on in-process cron. Tested with Vitest integration tests against real Postgres and Redis.

IoT-IDS: Edge Network Intrusion Detection and Prevention *2026 – Present*
Python, XGBoost, ONNX, scapy, C

- Streaming flow-based IDS/IPS that sniffs traffic, detects 9 attack types across 4 categories in a hierarchical taxonomy, aggregates incidents per source and can actively block offenders (IPS mode).
- Exported a 91.8 KB ONNX model with 8.1 μs single-flow inference on an Apple M4, and compiled the same model to a dependency-free C header ($\sim$43 KB) for microcontrollers.
- Built SFAF, a semantic feature-alignment framework over 11 public IDS datasets (including CICIDS2017, UNSW-NB15, TON-IoT and Bot-IoT), to measure cross-dataset transfer; the study surfaced severe domain shift between datasets.

Data Acquisition and Logging System for EV Testing *Apr 2024 – Apr 2025*
STM32 Blue Pill, Arduino, Raspberry Pi, RaceCapture, Lua, CAN, ADC, HC-05, Python

- Designed and built the data-acquisition system for a Formula Student electric car, with a driver interface, live telemetry, CAN-based communication, data logging and post-run processing.
- Collected real-time voltage, current, RPM, temperature, brake-pressure and suspension data over analog and digital channels, with Bluetooth telemetry and timestamped onboard SD logging.
- Wrote Python and shell tools for automated log parsing, visualization and performance mapping.

HR Tool for an Accounting Firm *2025*
React, TypeScript, Vite, Tailwind CSS, Supabase

- Built an internal HR tool with task management, leave tracking and payroll processing.

Leadership & Activities

Team Octane Racing Electric (Formula Student), COEP Pune, India
E-Powertrain Engineer & Data Acquisition Lead *Oct 2023 – May 2026*

- Developed an STM32-based ECU for the Blitz 1.0 (2025) car, bringing CAN communication, DMA and an RTOS to the team's car for the first time in its history; developed and tested ECU communication protocols.
- Led DAQ design for the Alectrona 3.0 (2024) car, including RaceCapture research and CAN integration.
- Worked on battery integration, motor-controller interfacing, high- and low-voltage safety systems, LV circuit testing and wiring-harness design; assisted in BMS assembly, including thermistor integration and subsystem validation.
- Researched and approached automotive companies for sponsorship, and represented the team at summits and conferences.
- Explored early-stage driverless perception with OpenCV (cone and track-boundary detection) and ROS2 for future competition entries.

Competitions

Formula Student Germany, Hockenheimring: Data Acquisition Engineer for COEP *Aug 2025*
Formula Bharat, Coimbatore: represented Team Octane Racing Electric and COEP *Jan 2024*

Certifications

NPTEL: Embedded Systems and RTOS *2025*
Coursera: ARM Cortex-M Microcontroller Programming *2025*
Udemy: STM32 Programming with HAL *2024*
Udemy: Driverless Car Engineering: Vision, Control and Planning *in progress*

Volunteer Experience

FLOS (For the Love Of Science), Wakad, Pune: media volunteer for an NGO running astronomy and space-observation camps in schools and housing societies *May 2024 – Jul 2024*

Skills

Languages: Python, C, C++, TypeScript/JavaScript, SQL, Java, Bash, Lua
Backend & Web: Node.js (Fastify), React, REST API design, PostgreSQL/TimescaleDB, Redis, Knex, Supabase, Docker, AWS (SES)
ML & Data: XGBoost, scikit-learn, PyTorch, pandas, ONNX, feature engineering, spatial cross-validation, Google Earth Engine, openEO
Embedded: STM32, ARM Cortex-M, ARM7, PIC, Arduino, Raspberry Pi, FreeRTOS, μC/OS-II, CAN, SPI, I²C, UART, DMA
Tools: Git, Linux, STM32CubeIDE, MATLAB/Simulink, LaTeX, AI coding assistants (Claude Code, Codex)
Spoken languages: English, Hindi, Marathi, German